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15 Implement Iris Flower Classification using Naive Bayes classifier

Program:

Iris Flower Classification using Naive Bayes

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix
```

Load Iris dataset

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

Split dataset into training and testing data

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=42
)
```

Create Naive Bayes classifier

```
model = GaussianNB()
```

Train the model

```
model.fit(X_train, y_train)
```

```
# Predict test data
```

```
y_pred = model.predict(X_test)
```

```
# Display results
```

```
print("=====")
```

```
print(" IRIS FLOWER CLASSIFICATION")
```

```
print("=====")
```

```
print("\nActual Values :", y_test)
```

```
print("Predicted Values :", y_pred)
```

```
# Calculate accuracy
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("\nAccuracy:", accuracy)
```

```
print("Accuracy Percentage:", accuracy * 100, "%")
```

```
# Confusion Matrix
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
print("\nConfusion Matrix:")
```

```
print(cm)
```

```
# Test a new flower
```

```
new_flower = [[5.1, 3.5, 1.4, 0.2]]
```

```
prediction = model.predict(new_flower)
```

```
print("\nNew Flower Details")
```

```

print("-----")
print("Sepal Length : 5.1 cm")
print("Sepal Width : 3.5 cm")
print("Petal Length : 1.4 cm")
print("Petal Width : 0.2 cm")

print("\nPredicted Flower:",
      iris.target_names[prediction[0]])

```

output:

```

=====
      IRIS FLOWER CLASSIFICATION
=====

Actual Values      : [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0
0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]
Predicted Values   : [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 2 2 1 1 2 0 2 0 2 2 2 2 2 0 0
0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Accuracy: 0.9777777777777777
Accuracy Percentage: 97.77777777777777 %

Confusion Matrix:
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]

New Flower Details
-----
Sepal Length : 5.1 cm
Sepal Width  : 3.5 cm
Petal Length : 1.4 cm
Petal Width  : 0.2 cm

Predicted Flower: setosa

```